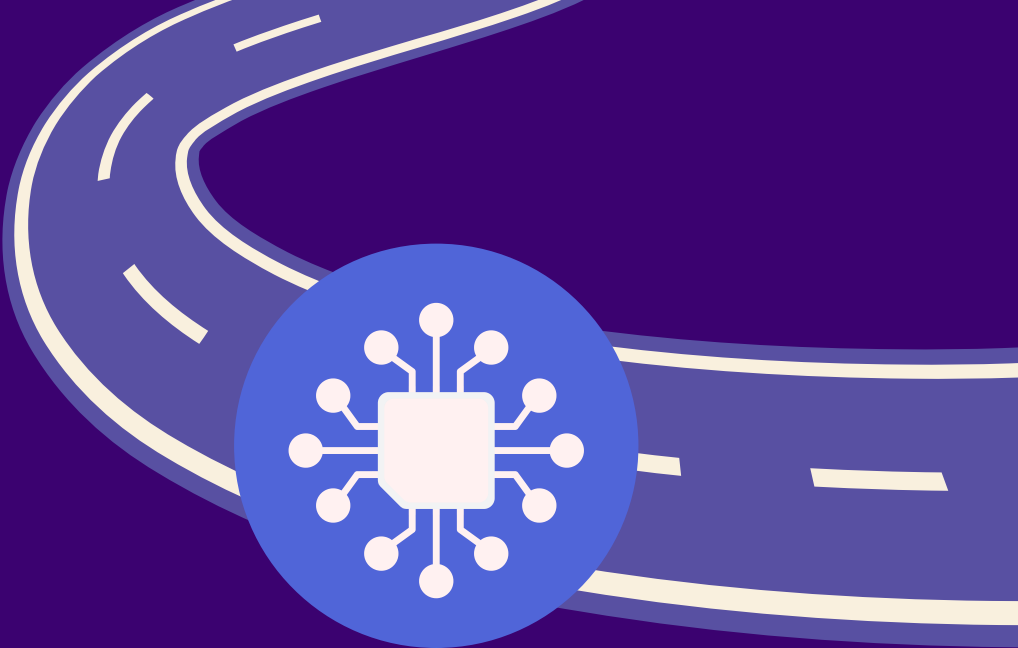




BUSGUARDIAN

" Every ride. Every child. Checked. "

PRESENTED BY:

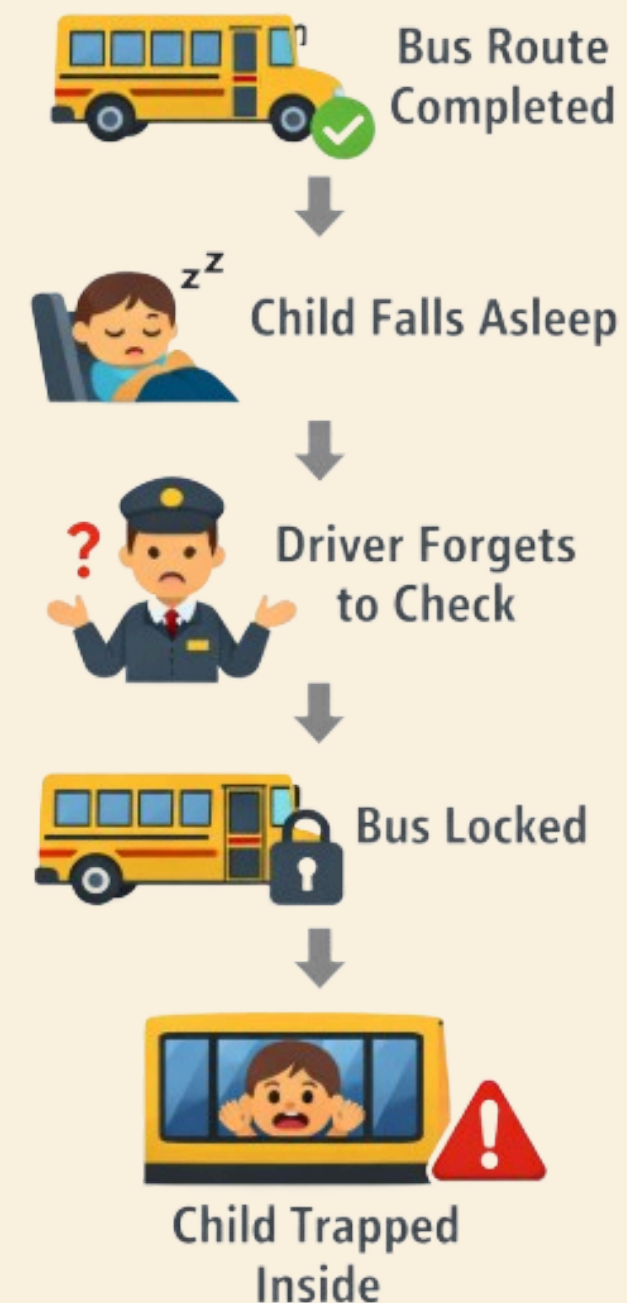
Name: Ananya Mishra

Role: Graduate Trainee Engineer



Problem :

- School buses carry many children every day.
- A child may fall asleep and remain inside after the route ends.
- Current safety checks rely on manual inspection.
- Human error can cause these checks to fail.
- An automated system is needed to detect a child inside the bus.
- The system should trigger immediate alerts to prevent harm.





My Solution :BusGaurdian

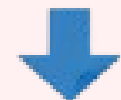
A Multi layer safety system- “BusGaurdian”



RFID Tracking
(Entry / Exit)



Seat Detection
(Seat Occupied?)



Driver Walk Check
(Rear Button Press)



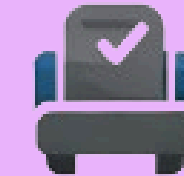
Alarm & Notifications
(Buzzer + SMS)

“Tech + Verification = No child left behind “

System Architecture



RFID Reader



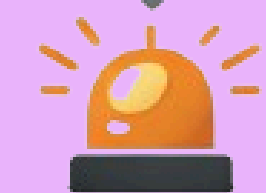
Seat Sensors



Rear Check Button



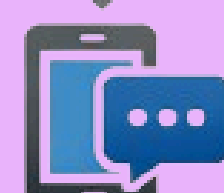
ESP32



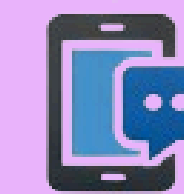
Buzzer Alarm



GSM Module



SMS Alert



“Central controller: Processes all safety signals”

Solution Description :

The system works using three levels of detection:

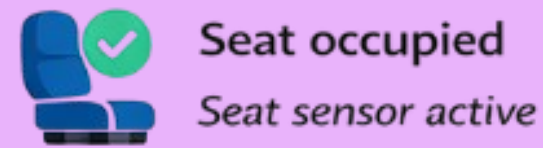
- **RFID Tracking:** Students scan their RFID cards when entering and exiting the bus. The system keeps track of the total passenger count.
- **Seat Detection:** Sensors installed under selected seats detect if any seat is still occupied after the route ends.
- **Driver Walk Check:** When the bus is parked, an alarm prompts the driver to walk to the rear of the bus and press a verification button.

“If a **child is detected** or a **passenger count mismatch** occurs, the system immediately **triggers a buzzer alarm** and **sends an SMS** alert to school authorities.”

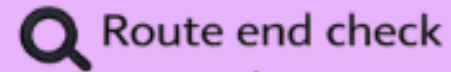
“This multi-layer verification approach reduces human error and ensures that every bus is checked before being locked.”

Working :

Seat Detection



Seat occupied
Seat sensor active



Route end check

NO

Bus Safe

YES

Possible

Detect child even if sleeping or inactive

Detection Logic



Route End



Check RFID Count



Driver walks to rear

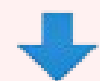


Press verification button

Entry/Exit Tracking



Child enters bus



RFID scan



Passenger count + 1

Entered = 30 ► Exited = 29

⚠ System detects mismatch

Driver Walk Check



Bus ignition OFF



Alarm activates



Driver walks to rear



Driver visually checks the entire bus.

Alert System



Child detected



Buzzer inside bus



SMS alert sent



School transport staff notified

Example message:

⚠ ALERT

Child detected in Bus #12
Immediate inspection required.

ESP32 Control Logic (System Logic)

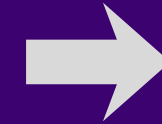
Entry / Exit Tracking Logic

- The RFID reader receives card scan data from students.
- ESP32 increases the entry count (+1) when a student boards the bus.
- When a student exits, the count decreases (-1).
- The system continuously maintains the current passenger count.



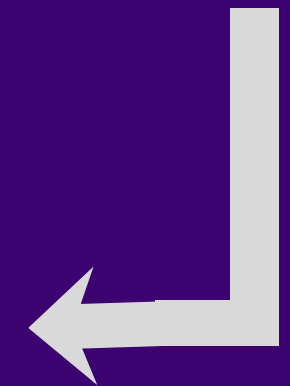
Route End Detection

- When the bus route ends:
- The system detects the ignition OFF signal, or
 - The driver presses the verification button.
- At this point, the system enters final safety check mode.



Seat Detection logic

- ESP32 reads data from the seat sensors.
- Seat occupied → TRUE
 - Seat empty → FALSE
- If any seat occupied = TRUE, the system suspects that a child may still be inside the bus.



Driver Walk Check Login

- The driver presses the rear verification button after checking the bus.
- If the system detects:
- RFID mismatch + seat occupied
- then driver confirmation is required before closing the inspection.



Alert Logic

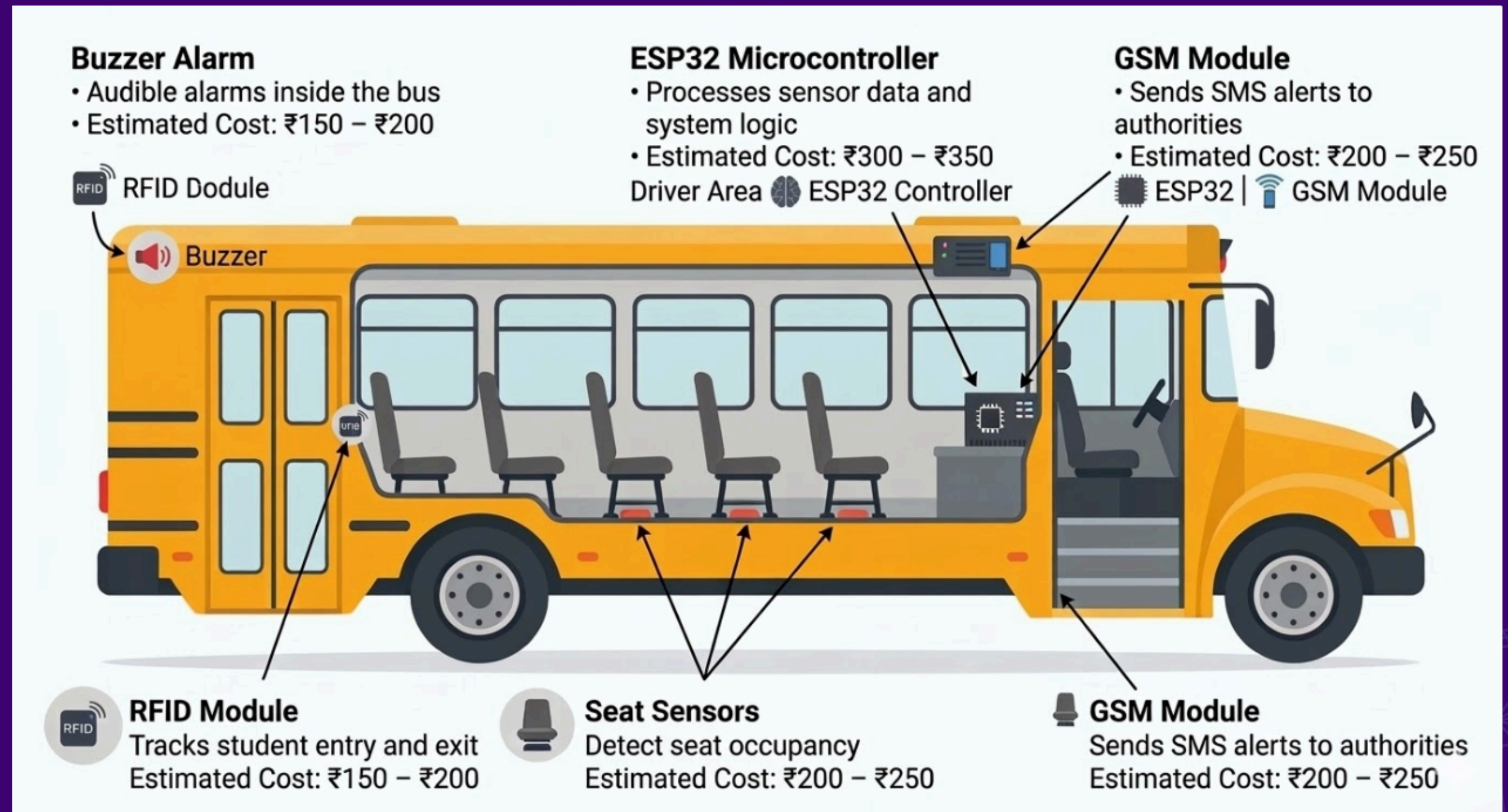
- If the system detects:
- Passenger count mismatch
 - OR
 - Seat occupancy after route completion
- ESP32 will:
- Turn ON the buzzer alarm inside the bus.
 - Trigger the GSM module to send an SMS alert.

Costing and Deployment :

Total Estimated Cost $\approx$ ₹800 – ₹1000 per bus

Estimated Installation time per bus $\approx$ 30 minutes per bus

*“This **low-cost setup** makes the solution **scalable** and **easy to deploy** across large school bus fleets.”*



Testing and Validation

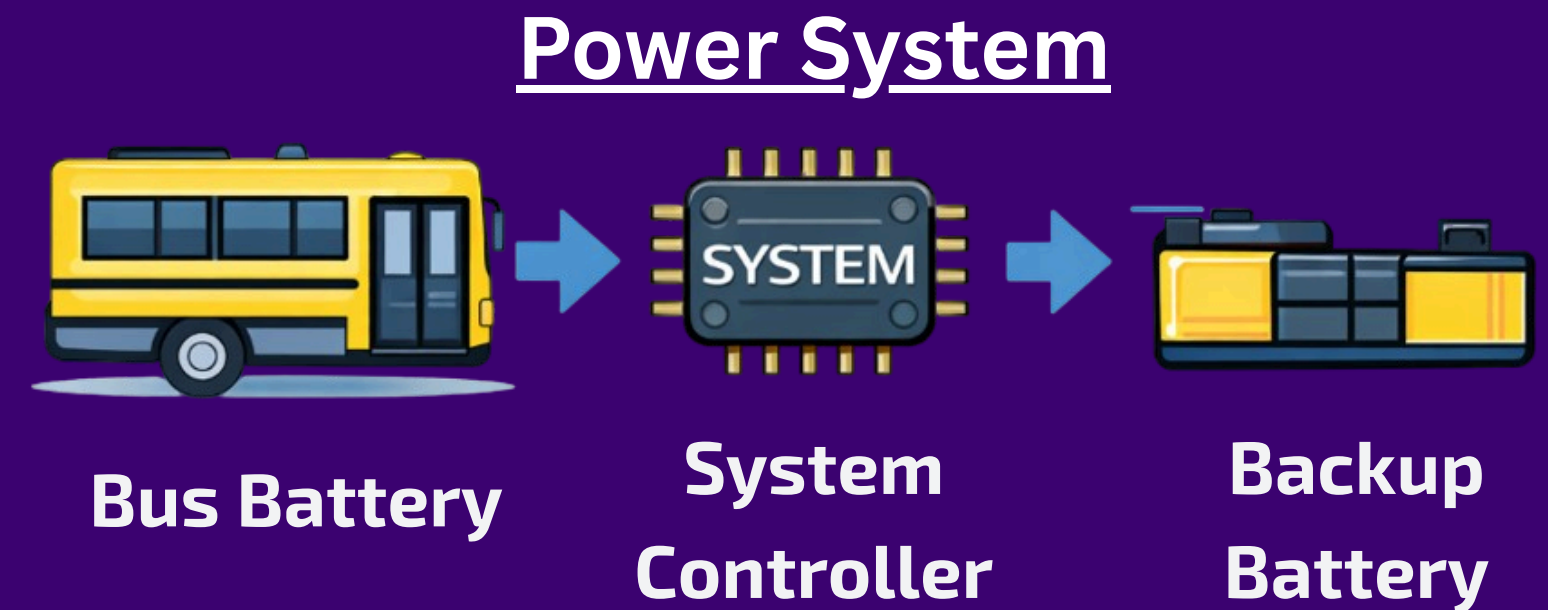
Test Scenarios

- 👤 Child sleeping → detected
- 📶 RFID mismatch → seat sensors verify
- 🚌 Empty bus → no alert
- 🚨 Seat occupied → alarm triggered

Validation Process

- Simulated different scenarios such as sleeping child and empty bus.
- Verified sensor response and alert triggering.
- Ensured the system works even when the child is inactive.

Safety & Operational Reliability



Environmental Reliability

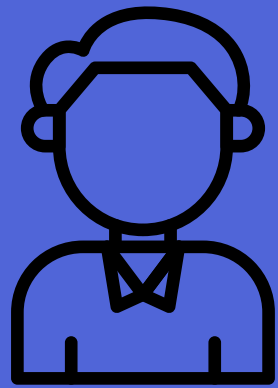
- 🌡️ High temperature
- ☁️ Dust & vibration
- 🌙 Low lighting

Privacy

- 🚫 No cameras
 - 🚫 No video recording
 - ✅ Only presence detection sensors
- Ensures child privacy & data safety.

“Works even after engine is off.”

Impact :



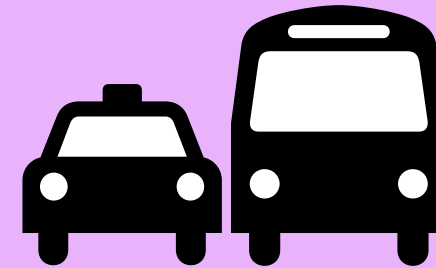
Child Safety

Ensures no child remains inside the bus after the trip.



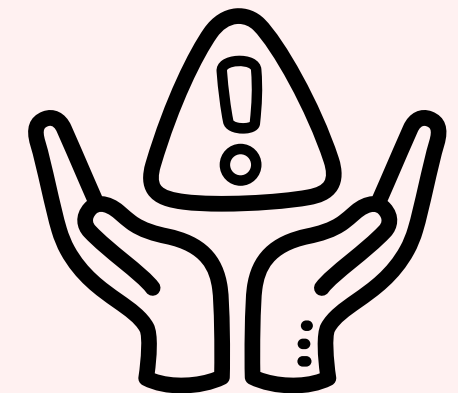
School Trust

Improves confidence of parents and schools in transport safety.



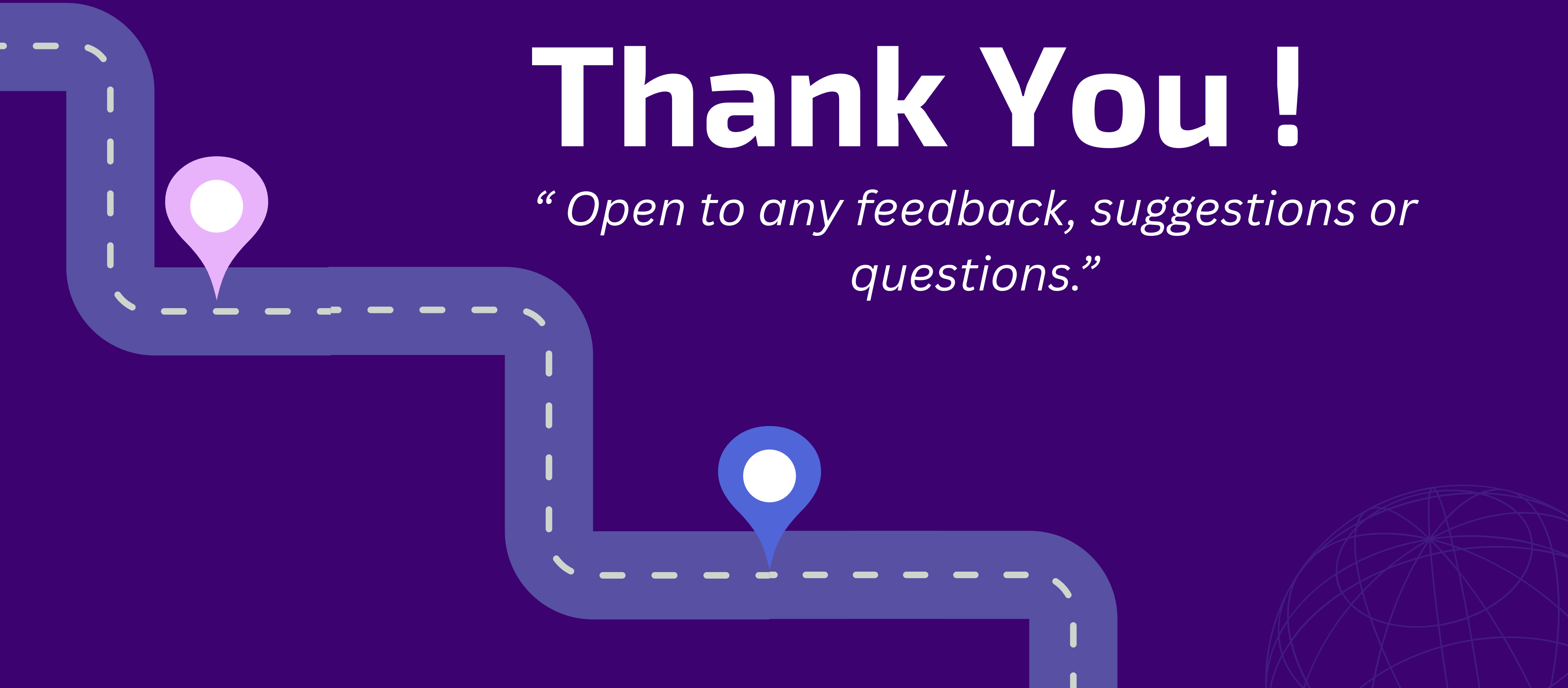
Safer Transport

Adds automated monitoring to school bus operations.



Reduced Risk

Minimizes accidents caused by missed manual inspections.



Thank You !

“Open to any feedback, suggestions or questions.”